



8/13/2026

ASSIGNMENT DAY 5

Case Study Practical: Troubleshooting
Multiple Hardware Issues.

TASK 1.

❖ A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

➤ Laptop Power-On Troubleshooting:

- **Check the Power Outlet:** Confirm that the outlet is working by testing another device.
- **Inspect the Charger:** Check the adapter, cable, and plug for visible damage or loose connections.
- **Check the Charging Port:** Look for dust, damage, or a loose connection in the laptop's power port.
- **Test the Charger:** Use a multimeter to check whether the charger is providing the correct output voltage.
- **Check the Battery:** If the charger works but the laptop still does not power on, the battery may be faulty.

TASK 2.

❖ List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

➤ Monitor Blank Screen Troubleshooting Flow:

- **Check Power:** Make sure the monitor is switched on and receiving power.
- **Check Display Cable:** Ensure the HDMI/DisplayPort cable is firmly connected to both the monitor and PC.
- **Check Input Source:** Select the correct HDMI/DisplayPort input on the monitor.
- **Check GPU Connection:** Make sure the graphics card is properly seated and connected.
- **Test the Display:** Try another cable or monitor to identify whether the problem is with the display hardware.
- **Check Display Settings:** If the display works, verify the correct display mode and resolution in Windows.
- **Confirm the Fix:** Restart the PC and check that the monitor displays the system normally.

TASK 3.

❖ **Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.**

➤ **Wireless Mouse Troubleshooting:**

- **Check the Mouse:** Replace the battery and make sure the mouse is switched on and properly paired with its receiver.
- **Test the USB Port:** Connect the mouse receiver to another USB port to check if the original port is faulty.

- **Test Another Mouse:** Connect a working mouse to the same USB port to identify whether the problem is with the mouse or the port.
- **Check System Settings:** Verify that the mouse is detected and enabled in the system settings.
- **Confirm the Cause:** If another mouse works but the wireless mouse does not, the original mouse or receiver is likely faulty.

TASK 4.

❖ **Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one. Think about fans, dust, and thermal paste.**

➤ **Gaming PC Overheating Hardware Checks:**

- **Faulty Fan:** Check whether the CPU/GPU fans are spinning properly and replace any faulty fan.
- **Dust Buildup:** Inspect the fans, heatsinks, and air vents for dust and clean them carefully.
- **Dry Thermal Paste:** Check the thermal paste between the CPU/GPU and heatsink; replace it if it is dry or damaged.